

Condition-Based Maintenance with Imperfect Preventive Repairs for a Deteriorating Production System

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This paper presents a condition-based maintenance policy considering aspects of maintenance cost and productivity for a single-unit deteriorating production system whose condition is periodically monitored. Imperfect preventive maintenance actions, which restore the production system to better than old, are considered with both deterministic and random intervention gains. Moreover, the system is assumed to be repaired only a limited number of times. In order to assess the performance of the proposed maintenance policy, different types of imperfect preventive maintenance cost functions are investigated. A numerical example of a multi-state production system is finally introduced to illustrate the use and the advantages of the proposed policy. Copyright © 2012 John Wiley & Sons, Ltd.

Keywords: imperfect maintenance; condition-based maintenance; production system; stochastic process; optimization

1. Introduction

Health monitoring equipments providing information about the system condition have evolved rapidly over the past decade. They can be considered as useful tools in maintenance decision-making. Based on the perfect or imperfect monitoring information, different condition-based or predictive maintenance policies which lead to avoid failure occurrence at the lowest cost can be performed, see for example.^{6–8,20,26} In the context of production system,^{1,9,24} the deterioration causes lower production rates (therefore higher production cost per item) and lower product quality. Preventive maintenance is one of the efficient tools to increase the reliability of the production system. Without an effective maintenance program, the system might be down for significant amounts of time. This means that the effective production rate decreases significantly and the system might not be able to cope with demand. In such maintenance policies, perfect preventive maintenance (or preventive replacement) is usually used. After each preventive maintenance action, the system is completely restored and considered as good as new. However, the replacement or perfect repair actions are often expensive. Imperfect maintenance relying on the fact that the system state after maintenance is somewhere between the state before maintenance and as good as new has grown recently as a popular issue to researchers, see for example.^{2,11,13,14,19,16} Various methods and optimal policies for imperfect maintenance are summarized and discussed by Pham and Wang.²¹ However, imperfect maintenance for condition-based maintenance of deteriorating systems has not received much attention and remains widely open.

The aim of this paper is to propose a condition-based maintenance policy for deteriorating production systems in which the evolution of condition (or degradation behavior) is assumed to be a Gamma stochastic process. The proposed maintenance policy considers imperfect preventive maintenance actions with both deterministic and random improvement gains. Besides, from a practical point of view, the system may not be repaired an infinite number of times due to technical or economical reasons, see for example.¹² In this paper, we assume that the system can be repaired only a limited number of times. Furthermore, in order to evaluate imperfect maintenance actions, an imperfect cost model with different types of preventive maintenance cost functions is investigated.

Moreover, in many preventive maintenance policies, the decision parameters (preventive threshold, inspection schedule, ...) are optimized based on only the maintenance cost per time unit without taking into consideration the productivity which may depend on the system condition, e.g. the degradation level, see.⁹ Maintenance cost per unit of product criterion, recently introduced in⁴ for a replacement policy, is developed in the present paper.

This paper is organized as follows. Section 2 is devoted to the description of the system characteristics and related assumptions. Section 3 focuses on the proposed imperfect maintenance policy for both deterministic and random intervention gains. The proposed

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imperfect maintenance cost model is also investigated. The evaluation and the optimization of the proposed maintenance policy are considered in Section 4. To illustrate the proposed maintenance policy, a simple numerical example is introduced in Section 5. Some numerical results are in addition discussed here. Finally, the last section reports the conclusions drawn from this work.

NOTATION LIST:

C_i	inspection cost
C_p^0	perfect (replacement) preventive maintenance cost
C_p^k	imperfect preventive maintenance cost of the k th maintenance action
C_c	corrective (replacement) maintenance cost
C_d	unavailability cost rate of the system
$d(t)$	total time passed in failed state in $[0, t]$
EC_∞	long-run expected maintenance cost rate
EP_∞	long-run expected productivity
ECP_∞	average of maintenance cost per unit of product
L_i	deterioration threshold level for operation mode (state) i
L, M	failure and preventive maintenance thresholds
m	system deterioration level after a deterministic intervention
N	number of operation modes of the system
N_i	number of inspection in $[0, t]$
N_p	number of preventive maintenance in $[0, t]$
N_c	number of corrective maintenance in $[0, t]$
N^{prev}	allowable imperfect maintenance number
T_i	T_i th inspection time
ΔT	inter-inspection length
S_i	production capacity of the system in operation mode (state) i
X_t	system deterioration level at time t
Z_{T_i}	intervention gain at the preventive maintenance time T_i
$u(.)$	degradation improvement factor
α, β	scale and shape parameters of deterioration process.

2. Model description and assumptions

For multi-state production systems such as manufacturing production lines and power generation installations, the performances output of interest is not only the availability but also the production capacity which can settle on different levels (e.g. 100%, 90%, 80% of the nominal capacity) depending on the system conditions.^{10,3,9,4}

2.1. General assumptions

We consider a single-unit system in which the condition (degradation level) at time t can be summarized by an observable random scalar variable X_t . This random variable can represent for example a crack length or defect products. In the absence of repair or replacement actions, the deterioration evolution of the system is assumed to be strictly increasing. The process $(X_t)_{t \geq 0}$ is then an increasing stochastic process. Moreover, we suppose that the following assumptions are verified.

- The initial state X_0 is 0
- The system is in failed if the ageing variable is greater than a level L . The threshold L can be seen as a deterioration level which must not be exceeded for economical or security reasons.
- In the allowed operation interval (i.e. from 0 to L), N different operation modes are considered. A unique production (or treatment) capacity S_i corresponds to each operation mode (or state) i according to the degradation level of the system. More precisely:
 - $X_t < L_1$: the system is in nominal state denoted state 1 in which it is functioning with nominal production capacity S_1
 - $L_1 \leq X_t < L_2$: the system is in operational state (state 2) in which its production capacity is $S_2 < S_1$.
 - ...
 - $L_{N-1} \leq X_t < L_N = L$: the system is still functioning but badly (state M), the production capacity for this case is $S_N < S_{N-1}$

The system degradation behavior and corresponding states are illustrated in Figure 1.

2.2. Deterioration modelling

Gamma processes have been widely used to describe the degradation of systems.^{26,8,25} A characteristic of this process is that it is clearly monotone increasing which is the behavior observed in most physical deterioration processes. Moreover, its paths are

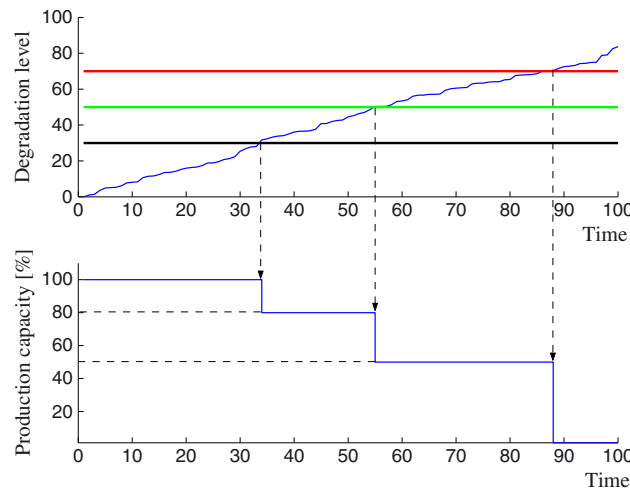


Figure 1. Illustration of degradation level and corresponding production capacities

discontinuous, and it can be thought as the accumulation of an infinite number of small shocks. Herein, the deterioration between two maintenance operations is assumed to evolve like a Gamma stochastic process $(\tilde{X}_t)_{t \geq 0}$, with the following characteristics:

- $\tilde{X}_0 = 0$;
- $(\tilde{X}_t)_{t \geq 0}$ has independent increments;
- for all $0 \leq l < t$, the random increment $\tilde{X}_t - \tilde{X}_l$ follows a Gamma probability density with shape parameter $\alpha(t-l)$ and scale parameter β :

$$f_{\beta, \alpha(t-l)}(x) = \frac{1}{\Gamma(\alpha(t-l))} \beta^{\alpha(t-l)} x^{\alpha(t-l)-1} e^{-\beta x} \mathcal{I}_{\{x \geq 0\}},$$

- where α is a positive real number and the indicator function $\mathcal{I}_{\{x \geq 0\}} = 1$ if $x \geq 0$, $\mathcal{I}_{\{x \geq 0\}} = 0$ otherwise.

The mean deterioration speed and its variance are α/β and α/β^2 , respectively. Various deterioration behaviors can be modeled by changing the couple of parameters α, β .

3. Maintenance policy

In the framework of condition-based maintenance, a maintenance policy relies essentially on two main decisions: when to take (preventively/correctively) maintenance actions and when to inspect. The system information (degradation level) is usually used to make the decision on the inspection time and on the maintenance action to perform. Herein, we assume that the system is periodically inspected with the inter-inspection interval length ΔT to be optimized. The i th inspection time of the system is determined by:

$$T_i = i\Delta T. \quad (1)$$

According to the degradation level X_{T_i} at inspection time T_i , the maintenance decision is the following:

- if $X_{T_i} \leq M$, the system is in a working state, no maintenance action is performed. M is called the preventive maintenance threshold, and it is a decision variable to be optimized.
- if $L > X_{T_i} > M$ (the system is still functioning but badly), an imperfect maintenance action is immediately carried out. This preventive maintenance action may not restore completely the system, but it can lead the system to be a better state at which the degradation level is lower or equal to the preventive maintenance threshold M . We assume, however that, within a life cycle (a life cycle starts when the system is new), the system can be repaired only a limited number of times N^{prev} , i.e. at $(N^{prev} + 1)$ th preventive maintenance, a perfect maintenance (or replacement) action is required. We assume that each perfect preventive maintenance costs C_p^0 .

Let $X_{T_i}^*$ be the degradation level of the system after maintenance; it is then calculated by:

$$X_{T_i}^* = X_{T_i} - Z_{T_i}. \quad (2)$$

where Z_{T_i} is the intervention gain. Both deterministic and random intervention gains are considered in the paper and will be presented later.

Each preventive action requires a maintenance cost C_p^k which may depend on both the intervention gain and the system condition at inspection time. The imperfect maintenance costs will be discussed more in detail in Section 2.

- if $X_{T_i} \geq L$, the system is failed, then a corrective replacement action is performed, and a cost C_c is incurred. An additional cost is incurred by the time d^j elapsed in the failed state at a unavailability cost rate C_d which may correspond to production loss per unit of time. After a corrective maintenance action, the system is considered as good as new (the degradation level after corrective maintenance is assumed to be 0).

We assume finally that maintenance and inspection durations are neglected.

3.1. Imperfect maintenance quality

It is shown in the literature that maintenance gain could be either deterministic, see for instance,²² or random,^{2,17} depending on different factors, e.g. the system characteristics, maintenance supports, ... In this way, both deterministic and random intervention gains are investigated in this work.

3.1.1. Deterministic interventions. We consider now the first case where imperfect maintenance action can be able to restore the system, if it is still functioning, to a better state with a given degradation level. We assume also that if an imperfect preventive maintenance action is performed at inspection time T_i , i.e. $X_{T_i} > M$, the desired degradation level of the system after maintenance is then m ($0 \leq m \leq M$); m is a decision variable of the policy, and it has to be optimized. Hence, the intervention gain need is then:

$$Z_{T_i} = X_{T_i} - m. \quad (3)$$

Figure 2 illustrates the evolution of the system degradation behavior and the maintenance policy with deterministic intervention gains. Another maintenance policy with random intervention gains is presented in the next section.

3.1.2. Random interventions. Another modelling option is to assume that if an imperfect preventive maintenance is performed at inspection time T_i , the intervention gain Z_{T_i} is then a continuous random variable. Assume next that thank to this maintenance action, the degradation level of the system after maintenance $X_{T_i}^*$ is somewhere between 0 and the degradation level before maintenance X_{T_i} , i.e. $0 \leq X_{T_i}^* \leq X_{T_i}$. Hence, according to equation (2), the intervention gain Z_{T_i} is random on the interval $[0, X_{T_i}]$. The illustration of the system degradation evolution and of the proposed maintenance policy is shown in Figure 3.

From a mathematical point of view, there are several kinds of continuous random variables²³ that can be used to model the intervention gain. Herein, uniform and truncated normal random variables are studied:

Uniform random intervention Consider that Z_{T_i} is described by a uniform random variable with density probability:

$$g_{a,b}^U(x) = \begin{cases} \frac{1}{b-a} & \text{if } a < x < b \\ 0 & \text{otherwise} \end{cases}$$

with,

- $a = 0$;
- $b = X_{T_i}$.

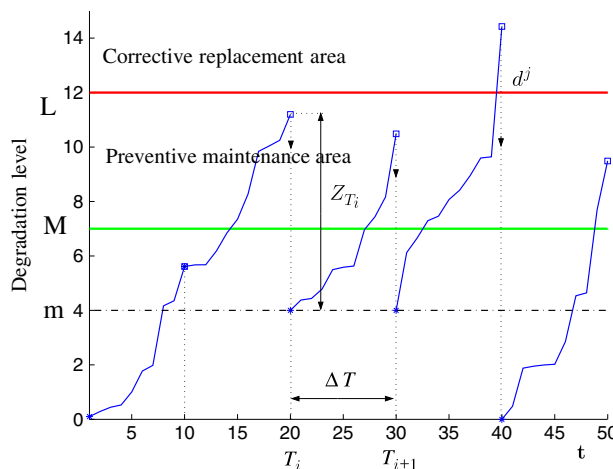


Figure 2. Illustration of degradation behavior and the proposed maintenance policy with deterministic intervention gains

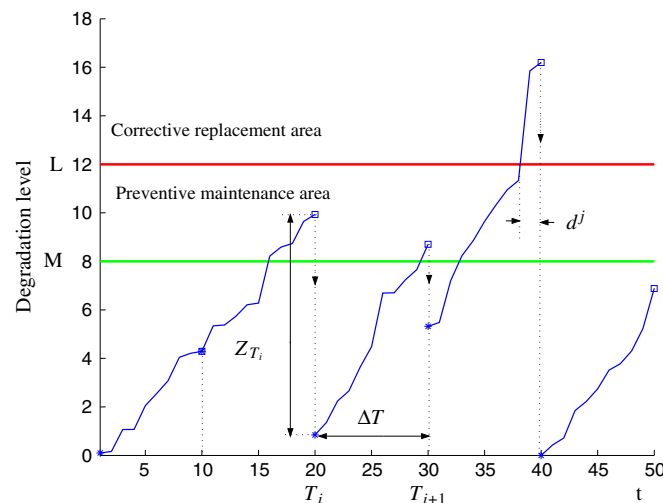


Figure 3. Illustration of degradation behavior and maintenance policy with random intervention gains

Truncated normal random intervention Assume now Z_{T_i} is distributed according to a truncated normal distribution with density:

$$g_{\mu, \sigma, a, b}^{TN}(x) = \frac{\frac{1}{\sigma} \phi\left(\frac{x-\mu}{\sigma}\right)}{\Phi\left(\frac{b-\mu}{\sigma}\right) - \Phi\left(\frac{a-\mu}{\sigma}\right)} \mathcal{I}_{[a, b]}(x), \quad (4)$$

where:

- $\mathcal{I}_{[a, b]}(x) = 1$ if $a \leq x \leq b$ and $\mathcal{I}_{[a, b]}(x) = 0$ otherwise;
- $\phi(\xi) = \frac{1}{\sqrt{2\pi}} \exp(-\frac{1}{2}\xi^2)$ is the probability density function of the standard normal distribution, and $\Phi(\cdot)$ is its cumulative distribution function;
- $\mu = X_{T_i}/2$ and $\sigma = X_{T_i}/6$;
- $a = \mu - 3\sigma = 0$ and $b = \mu + 3\sigma = X_{T_i}$.

According to this distribution, Z_{T_i} is restricted: $a \leq Z_{T_i} \leq b$ or $0 \leq Z_{T_i} \leq X_{T_i}$. The mean intervention gains is $\mathbb{E}(Z_{T_i}) = \mu$, and the variance is $\text{VAR}(Z_{T_i}) = 0.973\sigma^2$, see ²³.

3.2. Imperfect preventive maintenance cost

In the framework of imperfect maintenance strategies, the relationship between preventive maintenance cost and imperfect maintenance action may play an important role on maintenance decisions. Kijima *et al.* in ¹¹ introduces the age reduction factor, defined as the ratio of the system effective age after maintenance divided by the system effective age before maintenance, which seems to be very powerful for modelling imperfect maintenance actions. Different kinds of relationship between age reduction factor and imperfect maintenance cost are discussed in ^{1,16,22}.

In the framework of condition-based maintenance for deterioration systems, we propose here the degradation improvement factor resulting from an imperfect preventive action at inspection time T_i as follows:

$$u(T_i) = \frac{Z_{T_i}}{X_{T_i}}.$$

The degradation improvement factor $u(T_i)$ ($0 \leq u(T_i) \leq 1$) represents the maintenance quality (through a direct modelling of the maintenance effect on the deterioration level), and a bigger $u(T_i)$ means a greater improvement. When $u(T_i) = 1$, the maintenance action is perfect, and the system becomes as good as new. $u(T_i) = 0$ means that there is no improvement on the system degradation.

In general, each maintenance action incurs a cost, and an imperfect maintenance action often incurs a reduced maintenance cost, namely imperfect maintenance cost, which may be independent of maintenance quality, see for instance ^{2,13,17}. From a practical point of view, in most cases, the quality of the maintenance action increases with the level of resources allocated to it, and hence with its cost. As mentioned in ^{15,16,18} the age of the system and the spent maintenance cost are the two key factors that influence most the improvement factor which is herein regarded as $u(\cdot)$. If the spent maintenance cost approaches the replacement cost, the improvement would approach as good as new. Oppositely, if the spent maintenance cost is closed to zero, the improvement is closed to zero. On the other hand, the system age, which is regarded here as the system degradation level, is another important factor. A great improvement may cost a little when the system is young (when the system deterioration level is low). However, as a system ages (when the system deterioration level is high), improvement may be smaller, even when a large maintenance cost is allocated.

In this paper, when an imperfect preventive maintenance is performed at inspection time T_i , we have to pay a maintenance cost which is defined as:

$$C_p^k = C_p^0 \cdot u(T_i)^\eta, \quad (5)$$

where η is a positive real number and represents the imperfect preventive maintenance characteristic of the system. According to this model, different kinds of maintenance cost function can be found depending on the value of η . More precisely:

- when $\eta < 1$, imperfect maintenance cost C_p^k is a concave function: the maintenance cost increases more than the improvement gain when performing the maintenance;
- when $\eta = 1$, C_p^k is a linear function which implies that the maintenance cost is proportional to the improvement level gain;
- when $\eta > 1$, C_p^k is a convex function: the maintenance cost increases less than the improvement gain.

Figure 4 illustrates these three different shapes of the imperfect maintenance cost function. A case study of the proposed imperfect maintenance policy with different kinds of imperfect maintenance cost functions will be presented in the next section.

4. Optimization and evaluation of the maintenance policy

In maintenance optimization, maintenance cost per unit of time is usually used as a relevant performance criterion, see for example.^{8,27,25} However, in the framework of production system, productivity takes an important role and should be taken into account in the optimization procedure. In this paper, both maintenance cost and productivity are considered in the objective function.

The cumulative maintenance cost at time t is

$$C(t) = C_i \cdot N_i(t) + \sum_{k=1}^{N_p(t)} C_p^k + \sum_{j=1}^{N_c(t)} C_c + C_d \cdot d(t), \quad (6)$$

where:

- $N_i(t)$ is the number of inspection in $[0, t]$

$$N_i(t) = \sum_{n \in \mathcal{N}} \mathcal{I}_{\{t \leq T_n\}}$$

- $N_p(t)$ is the number of preventive maintenance in $[0, t]$

$$N_p(t) = \sum_{n \in \mathcal{N}} \mathcal{I}_{\{t \leq T_n\}} \mathcal{I}_{\{M < X_{T_n} < L\}}$$

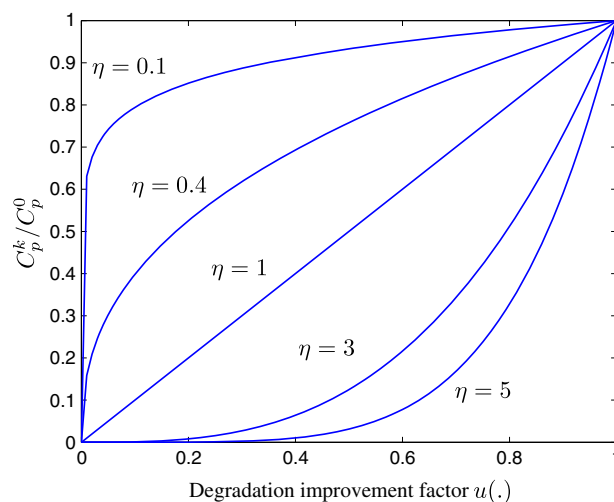


Figure 4. Illustration of imperfect maintenance cost function

- $N_c(t)$ is the number of corrective replacement in $[0, t]$

$$N_c(t) = \sum_{n \in \mathcal{N}} \mathcal{I}_{\{t \leq T_n\}} \mathcal{I}_{\{X_{T_n} > L\}}$$

- $d(t)$ is the total time passed in a failed state in $[0, t]$

$$d(t) = \sum_{j=1}^{N_c(t)} d^j = \int_0^t \mathcal{I}_{\{X_l \geq L\}} dl.$$

The long-run expected cost rate is:

$$EC_\infty = \lim_{t \rightarrow \infty} \frac{E[C(t)]}{t}. \quad (7)$$

The cumulative production capacity at time t is:

$$P(t) = \sum_{i=1}^N D_i(t) \cdot S_i, \quad (8)$$

where $D_i(t)$ is time passed in state i in the interval time $[0, t]$.

$$D_i(t) = \int_0^t \mathcal{I}_{\{L_{i-1} < X_l < L_i\}} dl.$$

In the same manner, the long-run expected productivity is:

$$EP_\infty = \lim_{t \rightarrow \infty} \left[\frac{E[P(t)]}{t} \right]. \quad (9)$$

The average of maintenance cost per unit of product can be defined as the ratio of the long-run expected cost rate divided by the long-run expected productivity rate.

$$ECP_\infty = \frac{EC_\infty}{EP_\infty}. \quad (10)$$

This quantity is used as the objective function to be minimized to find the optimal values of the decision variables in the proposed maintenance policy. To evaluate the objective function, Stochastic Monte Carlo simulation approach is used.

5. Numerical example

The purpose of this section is to show how the proposed maintenance model can be used in maintenance optimization of production deteriorating systems through a simple example.

Consider a single-unit production system in which its degradation behavior is assumed to be described by a Gamma process with scale parameter $\alpha = 1$ and shape parameter $\beta = 5$. The nominal production capacity of the system is 100 products/unit of time ($S_{nominal} = 100$). The production capacity can be changed according to the degradation level of the system. More precisely:

- when the degradation level $X_t < 20$, the system is in state 1 (normal operation state), its production capacity is $S_1 = S_{nominal} = 100$;
- $20 \leq X_t < 40$, the system is in state 2, its production capacity is $S_2 = 60$;
- $40 \leq X_t < 60$ (state 3), the production capacity is $S_3 = 40$;
- and when $60 \leq X_t$ (system fails), the production capacity is $S_4 = 0$.

Assume that inspection, preventive, corrective replacement costs and unavailability cost rate are respectively $C_i = 20$, $C_p = 90$, $C_c = 100$, $C_d = 100$ (all costs are given in arbitrary units). It is also assumed that the system can be preventively repaired only 10 times ($N^{prev} = 10$) for a life cycle. In both deterministic and random intervention cases, the imperfection maintenance costs shown in equation (5) are used with $\eta = 0.1, 0.4, 1, 3, 5$, i.e. three types of cost functions (concave, linear and concave forms) are considered. To evaluate the expected total maintenance cost and the expected productivity, the simulations are done on a very large interval of time so that t is assumed to tend infinity.

5.1. Deterministic interventions policy

Consider the first case where intervention gains are deterministic. To find the optimal value of the decision parameters (ΔT , M and m), the average of maintenance cost per unit of product is first evaluated with different values of ΔT , M and m . A numerical optimization scheme is secondly used. Table I reports the optimum values of ΔT and M and the minimum values of ECP_{∞} for different imperfect maintenance cost functions ($\eta = 0.1, 0.4, 1, 3, 5$).

The results show that when $\eta \leq 1$, the maintenance cost remains unchanged, the optimal maintenance policies correspond to a perfect maintenance policy ($m^{op} = 0$) where the degradation improvement factor is equal to 1. When η is higher than 1, the higher the η is, the lower the average of maintenance cost per unit of product is. Moreover, to compare with a perfect maintenance policy, the residual useful life-based maintenance policy,^{4,8} which seems to be an efficient policy in the framework of perfect condition-based maintenance, has been used and, with this policy, the minimum average maintenance cost per unit of product is 0.1869. It is clear that, according to the maintenance cost per unit of product, this perfect maintenance policy is better than the proposed imperfect maintenance policy only when $\eta \leq 1$. This interesting results may be useful for maintenance managers to choose the optimal maintenance strategy.

5.2. Random interventions policy

Consider now the second case with random interventions, i.e. the intervention gains can be described by uniform or truncated normal random distributions, see again paragraph 2. In the same manner, the optimal decision parameters can be reached by using a numerical optimization scheme and the obtained results for uniform and truncated normal random intervention gains are reported in Tables II and III, respectively.

According to the results, it is shown that in both cases (uniform and truncated normal random intervention gains), the higher the η is, the lower the average of maintenance cost per unit of product is. When $\eta \leq 1$, the imperfect maintenance actions are costly, perfect maintenance may be a better choice. However, when $\eta > 1$ (the imperfect maintenance cost function is a convex function with respect to the improvement factor), especially when $\eta = 5$, the proposed imperfect policy is much more cheaper than a perfect one and seems to be a better choice.

Table I. Imperfect preventive maintenance strategies with deterministic intervention gains

Optimal decision parameters						
η	ΔT^{op}	M^{op}	m^{op}	ECP_{∞}^{min}	EC_{∞}	EP_{∞}
0.1	8	18	0	0.1984	15.28	77.02
0.4	8	18	0	0.1984	15.28	77.02
1	8	18	0	0.1984	15.28	77.02
3	5	14	13	0.1504	10.53	70.01
5	5	15	12	0.1172	8.42	71.84

Table II. Imperfect preventive maintenance strategies with uniform random intervention gains

Optimal decision parameters					
η	ΔT^{op}	M^{op}	ECP_{∞}^{min}	EC_{∞}	EP_{∞}
0.1	4	28	0.3585	23.05	69.87
0.4	4	26	0.3167	20.67	65.27
1	4	19	0.2677	18.32	68.43
3	4	17	0.2037	14.25	69.96
5	3	16	0.1804	13.64	75.61

Table III. Imperfect preventive maintenance strategies with truncated normal random intervention gains

Optimal decision parameters					
η	ΔT^{op}	M^{op}	ECP_{∞}^{min}	EC_{∞}	EP_{∞}
0.1	5	31	0.3608	21.40	59.31
0.4	4	30	0.3244	20.07	61.86
1	4	22	0.2630	17.25	65.60
3	4	18	0.1731	11.64	67.24
5	3	12	0.1437	9.89	68.82

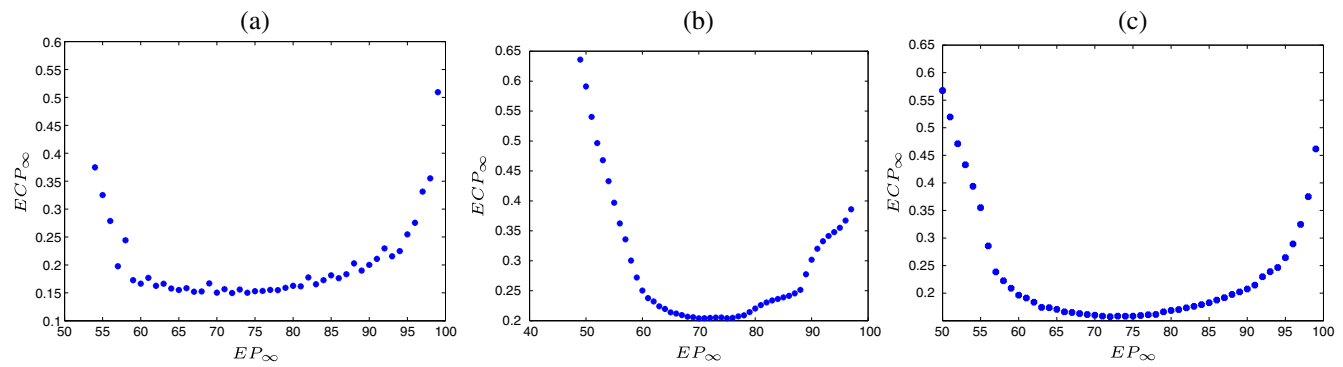


Figure 5. Maintenance cost per unit of product as a function of productivity for the case $\eta = 3$: (a) deterministic intervention gains; (b) uniform random intervention gains; (c) truncated normal random intervention gains

The results show also that the nature of intervention gains has a significant influence on the performance of the imperfect maintenance policy. With the same imperfect maintenance cost function, the random intervention gains are more costly than the deterministic intervention gain. This can be explained by the fact that in the deterministic intervention case, we can control accurately the desired state (degradation level) of the system after maintenance which is optimally obtained with the maintenance lowest cost. For the random interventions, the system state after maintenance remains random, which may lead the system to fail quickly after a maintenance operation, consequently, it may lead to a higher maintenance cost.

5.3. Optimal maintenance policy for a given productivity level

In industrial engineering, the production planning is often established according to the system capacity and the customer demands. It is therefore interesting to find an optimal maintenance policy when the system is required to operate with a given productivity level.

To illustrate this point, a non-linear imperfect maintenance cost function, $\eta = 3$, is used. The relationship between the minimum average of maintenance cost per unit of product and the productivity for both deterministic and random intervention gains are sketched in Figure 5. Each point corresponds to an optimal maintenance policy represented by the optimal decision parameters $\Delta T^{op}, M^{op}$ (and m^{op} for deterministic intervention case). Based on these results, the optimal maintenance policy can be found for a given productivity.

6. Conclusions

In this work, a condition-based maintenance under imperfect repairs with deterministic/random intervention gains for a deteriorating production system is described. Both cost and productivity are taken into account in the optimization procedure. This can help to find an adaptive optimal decision under a production level demand. The results show that the efficiency of imperfect maintenance policies may depend on the nature of intervention gains and on their related cost which is considered as a function of the intervention gains. When compared to the residual useful life-based maintenance policy, the proposed imperfect maintenance policy can provide better results especially when the imperfect maintenance cost function is a convex one.

This paper is the development of our research in the framework of condition-based maintenance with imperfect repairs for a deteriorating production system presented in part in.⁵ Our future research work will focus on the mathematical development in order to evaluate analytically the objective function. Furthermore, the proposed imperfect maintenance policy will be also developed for multi-unit systems with different degradation behaviors.

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